CareerFoundry ML with Python
December 27, 2025
Charlotte Lin

Achievement 2.2:
Complex ML Models and Keras Part 1

In this exercise, a convolutional neural network (CNN) was developed to classify “pleasant weather” conditions across the ClimateWins European weather stations using a multi-label learning framework. The analysis focused on preparing the data in the correct shape for deep learning, building and training a baseline CNN model, and evaluating performance using loss, accuracy, AUC, and per-label confusion metrics. Model behavior was assessed through learning curves and per-station performance, followed by targeted hyperparameter tuning to improve generalization and classification balance across stations.

Starting hyperparameters

```
Starting hyperparameters:
{'name': 'cnn_start', 'filters1': 16, 'kernel1': 2, 'act1': 'relu', 'pool1': 2, 'dropout
1': 0.1, 'filters2': None, 'kernel2': 2, 'act2': 'relu', 'pool2': 2, 'dropout2': 0.0, 'den
se1': 32, 'dense_act1': 'relu', 'dropout_dense': 0.2, 'batchnorm': False, 'lr': 0.001, 'ba
tch_size': 32, 'epochs': 8}
```

Final hyperparameters

```
Tuned hyperparameters:
{'name': 'cnn_tuned', 'filters1': 32, 'kernel1': 2, 'act1': 'relu', 'pool1': 2, 'dropout
1': 0.15, 'filters2': 64, 'kernel2': 2, 'act2': 'relu', 'pool2': 2, 'dropout2': 0.15, 'den
se1': 64, 'dense_act1': 'relu', 'dropout_dense': 0.25, 'batchnorm': True, 'lr': 0.0005, 'b
atch_size': 32, 'epochs': 12}
```

Hyperparameter tuning focused on increasing model capacity and stability by adding a second convolutional block, increasing filter counts, introducing batch normalization, and slightly reducing the learning rate. These changes were guided by baseline learning curves and resulted in improved validation loss, AUC, and macro-F1 without signs of overfitting.

Starting model summary

```
Model: "cnn_start"
_________________________________________________________________
 Layer (type)                Output Shape              Param #
=================================================================
 conv1d_1 (Conv1D)           (None, 14, 16)            304

 max_pooling1d_1 (MaxPoolin  (None, 7, 16)             0
 g1D)

 dropout_2 (Dropout)         (None, 7, 16)             0

 flatten_1 (Flatten)         (None, 112)               0

 dense_1 (Dense)             (None, 32)                3616

 dropout_3 (Dropout)         (None, 32)                0

 output (Dense)              (None, 15)                495

=================================================================
Total params: 4415 (17.25 KB)
Trainable params: 4415 (17.25 KB)
Non-trainable params: 0 (0.00 Byte)
```

```
Epoch 1/8
431/431 - 1s - loss: 0.4209 - bin_acc: 0.7963 - auc: 0.8323 - val_loss: 0.2826 - val_bin_acc: 0.8662 - val_auc: 0.9229 - 942ms/epoch - 2ms/step
Epoch 2/8
431/431 - 0s - loss: 0.2935 - bin_acc: 0.8582 - auc: 0.9156 - val_loss: 0.2568 - val_bin_acc: 0.8827 - val_auc: 0.9383 - 250ms/epoch - 581us/step
Epoch 3/8
431/431 - 0s - loss: 0.2700 - bin_acc: 0.8733 - auc: 0.9300 - val_loss: 0.2329 - val_bin_acc: 0.8930 - val_auc: 0.9498 - 262ms/epoch - 608us/step
Epoch 4/8
431/431 - 0s - loss: 0.2537 - bin_acc: 0.8810 - auc: 0.9387 - val_loss: 0.2245 - val_bin_acc: 0.8984 - val_auc: 0.9540 - 250ms/epoch - 579us/step
Epoch 5/8
431/431 - 0s - loss: 0.2457 - bin_acc: 0.8861 - auc: 0.9429 - val_loss: 0.2174 - val_bin_acc: 0.9017 - val_auc: 0.9576 - 255ms/epoch - 593us/step
Epoch 6/8
431/431 - 0s - loss: 0.2383 - bin_acc: 0.8897 - auc: 0.9465 - val_loss: 0.2115 - val_bin_acc: 0.9043 - val_auc: 0.9585 - 249ms/epoch - 578us/step
Epoch 7/8
431/431 - 0s - loss: 0.2332 - bin_acc: 0.8932 - auc: 0.9488 - val_loss: 0.2066 - val_bin_acc: 0.9075 - val_auc: 0.9606 - 250ms/epoch - 579us/step
Epoch 8/8
431/431 - 0s - loss: 0.2304 - bin_acc: 0.8938 - auc: 0.9502 - val_loss: 0.2025 - val_bin_acc: 0.9101 - val_auc: 0.9627 - 270ms/epoch - 626us/step
```

Tuned model summary

```
Model: "cnn_tuned"
_________________________________________________________________
 Layer (type)                Output Shape              Param #   
=================================================================
 conv1d_2 (Conv1D)           (None, 14, 32)            608       
                                                                 
 batch_normalization (Batch  (None, 14, 32)            128       
 Normalization)                                                  
                                                                 
 max_pooling1d_2 (MaxPoolin  (None, 7, 32)             0         
 g1D)                                                            
                                                                 
 dropout_4 (Dropout)         (None, 7, 32)             0         
                                                                 
 conv1d_3 (Conv1D)           (None, 6, 64)             4160      
                                                                 
 batch_normalization_1 (Bat  (None, 6, 64)             256       
 chNormalization)                                                
                                                                 
 max_pooling1d_3 (MaxPoolin  (None, 3, 64)             0         
 g1D)                                                            
                                                                 
 dropout_5 (Dropout)         (None, 3, 64)             0         
                                                                 
 flatten_2 (Flatten)         (None, 192)               0         
                                                                 
 dense_2 (Dense)             (None, 64)                12352     
                                                                 
 dropout_6 (Dropout)         (None, 64)                0         
                                                                 
 output (Dense)              (None, 15)                975       
                                                                 
=================================================================
Total params: 18479 (72.18 KB)
Trainable params: 18287 (71.43 KB)
Non-trainable params: 192 (768.00 Byte)
```

```
Epoch 1/12
431/431 - 1s - loss: 0.3404 - bin_acc: 0.8327 - auc: 0.8863 - val_loss: 0.2432 - val_bin_acc: 0.8885 - val_auc: 0.9462 - 1s/epoch - 3ms/step
Epoch 2/12
431/431 - 1s - loss: 0.2641 - bin_acc: 0.8760 - auc: 0.9334 - val_loss: 0.2188 - val_bin_acc: 0.9006 - val_auc: 0.9563 - 640ms/epoch - 1ms/step
Epoch 3/12
431/431 - 1s - loss: 0.2454 - bin_acc: 0.8871 - auc: 0.9432 - val_loss: 0.2179 - val_bin_acc: 0.8990 - val_auc: 0.9598 - 603ms/epoch - 1ms/step
Epoch 4/12
431/431 - 1s - loss: 0.2353 - bin_acc: 0.8912 - auc: 0.9481 - val_loss: 0.2002 - val_bin_acc: 0.9083 - val_auc: 0.9633 - 610ms/epoch - 1ms/step
Epoch 5/12
431/431 - 1s - loss: 0.2269 - bin_acc: 0.8961 - auc: 0.9519 - val_loss: 0.1991 - val_bin_acc: 0.9105 - val_auc: 0.9659 - 768ms/epoch - 2ms/step
Epoch 6/12
431/431 - 1s - loss: 0.2197 - bin_acc: 0.9002 - auc: 0.9550 - val_loss: 0.1900 - val_bin_acc: 0.9145 - val_auc: 0.9674 - 641ms/epoch - 1ms/step
Epoch 7/12
431/431 - 1s - loss: 0.2146 - bin_acc: 0.9027 - auc: 0.9572 - val_loss: 0.1809 - val_bin_acc: 0.9199 - val_auc: 0.9704 - 741ms/epoch - 2ms/step
Epoch 8/12
431/431 - 1s - loss: 0.2103 - bin_acc: 0.9052 - auc: 0.9591 - val_loss: 0.1783 - val_bin_acc: 0.9222 - val_auc: 0.9718 - 682ms/epoch - 2ms/step
Epoch 9/12
431/431 - 1s - loss: 0.2048 - bin_acc: 0.9078 - auc: 0.9612 - val_loss: 0.1834 - val_bin_acc: 0.9185 - val_auc: 0.9729 - 685ms/epoch - 2ms/step
Epoch 10/12
431/431 - 1s - loss: 0.2007 - bin_acc: 0.9112 - auc: 0.9628 - val_loss: 0.1715 - val_bin_acc: 0.9255 - val_auc: 0.9747 - 833ms/epoch - 2ms/step
Epoch 11/12
431/431 - 1s - loss: 0.1956 - bin_acc: 0.9132 - auc: 0.9648 - val_loss: 0.1646 - val_bin_acc: 0.9279 - val_auc: 0.9755 - 712ms/epoch - 2ms/step
Epoch 12/12
431/431 - 1s - loss: 0.1916 - bin_acc: 0.9151 - auc: 0.9663 - val_loss: 0.1611 - val_bin_acc: 0.9303 - val_auc: 0.9774 - 665ms/epoch - 2ms/step

TUNED macro metrics: {'macro_precision': 0.7968581353775814, 'macro_recall': 0.7001745083647695, 'macro_f1': 0.7368351899152809, 'label_cardinality_t
rue': 3.173753921226908, 'label_cardinality_pred': 2.9445799930289303}
```

Table of starting per-label confusion metrics (TP/FP/TN/FN + F1)

	label	support_pos	TP	FP	TN	FN	precision	recall	f1
9	MADRID_pleasant_weather	2570	2326	288	2880	244	0.889824	0.905058	0.897377
1	BELGRADE_pleasant_weather	1962	1675	435	3341	287	0.793839	0.853721	0.822692
2	BUDAPEST_pleasant_weather	1838	1532	387	3513	306	0.798332	0.833515	0.815544
7	LJUBLJANA_pleasant_weather	1543	1240	320	3875	303	0.794872	0.803629	0.799227
8	MAASTRICHT_pleasant_weather	1176	878	185	4377	298	0.825964	0.746599	0.784279
4	DUSSELDORF_pleasant_weather	1231	905	197	4310	326	0.821234	0.735175	0.775825
3	DEBILT_pleasant_weather	1101	810	204	4433	291	0.798817	0.735695	0.765957
13	STOCKHOLM_pleasant_weather	972	766	274	4492	206	0.736538	0.788066	0.761431
11	OSLO_pleasant_weather	859	604	178	4701	255	0.772379	0.703143	0.736137
10	MUNCHENB_pleasant_weather	1192	832	257	4289	360	0.764004	0.697987	0.729505
0	BASEL_pleasant_weather	1400	907	320	4018	493	0.739201	0.647857	0.690522
5	HEATHROW_pleasant_weather	1168	763	307	4263	405	0.713084	0.653253	0.681859
6	KASSEL_pleasant_weather	923	578	208	4607	345	0.735369	0.626219	0.676419
14	VALENTIA_pleasant_weather	276	2	0	5462	274	1.000000	0.007246	0.014388
12	SONNBLICK_pleasant_weather	0	0	0	5738	0	0.000000	0.000000	0.000000

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Table of tuned per-label confusion metrics

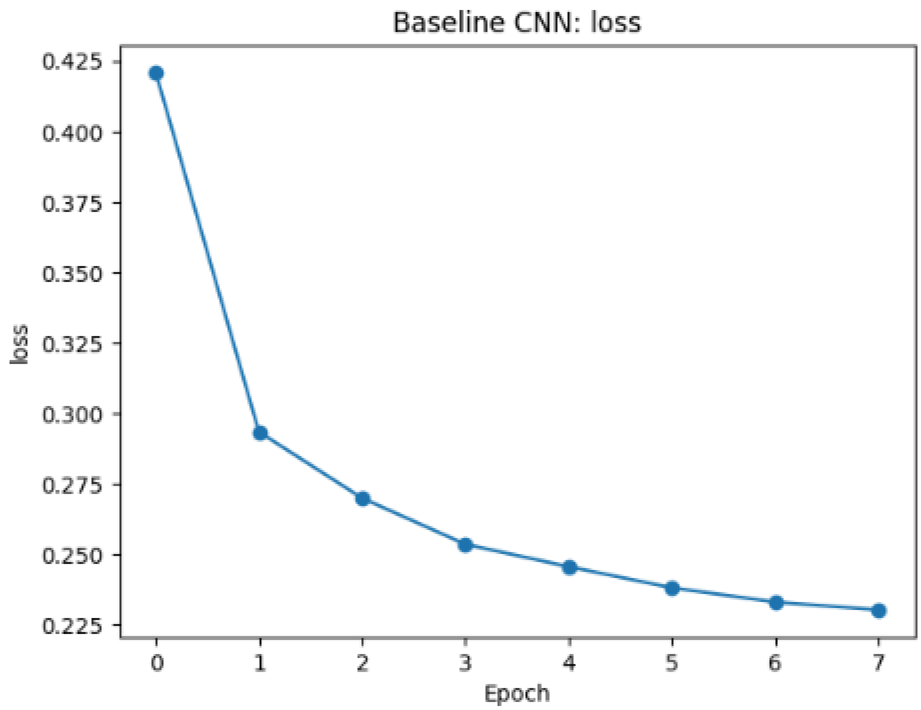
	label	support_pos	TP	FP	TN	FN	precision	recall	f1
9	MADRID_pleasant_weather	2570	2396	238	2930	174	0.909643	0.932296	0.920830
1	BELGRADE_pleasant_weather	1962	1610	211	3565	352	0.884130	0.820591	0.851176
2	BUDAPEST_pleasant_weather	1838	1541	243	3657	297	0.863789	0.838411	0.850911
7	LJUBLJANA_pleasant_weather	1543	1319	256	3939	224	0.837460	0.854828	0.846055
13	STOCKHOLM_pleasant_weather	972	787	129	4637	185	0.859170	0.809671	0.833686
10	MUNCHENB_pleasant_weather	1192	919	153	4393	273	0.857276	0.770973	0.811837
8	MAASTRICHT_pleasant_weather	1176	905	155	4407	271	0.853774	0.769558	0.809481
4	DUSSELDORF_pleasant_weather	1231	945	164	4343	286	0.852119	0.767669	0.807692
11	OSLO_pleasant_weather	859	656	119	4760	203	0.846452	0.763679	0.802938
3	DEBILT_pleasant_weather	1101	814	140	4497	287	0.853249	0.739328	0.792214
0	BASEL_pleasant_weather	1400	1017	240	4098	383	0.809069	0.726429	0.765525
6	KASSEL_pleasant_weather	923	608	80	4735	315	0.883721	0.658722	0.754811
5	HEATHROW_pleasant_weather	1168	863	292	4278	305	0.747186	0.738870	0.743005
14	VALENTIA_pleasant_weather	276	86	10	5452	190	0.895833	0.311594	0.462366
12	SONNBLICK_pleasant_weather	0	0	0	5738	0	0.000000	0.000000	0.000000

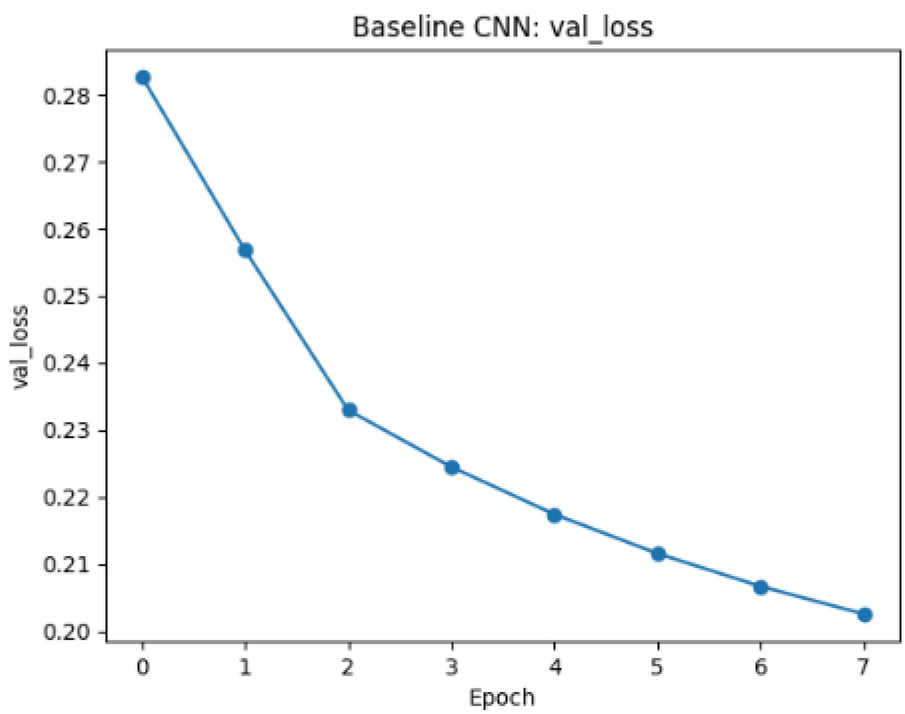
Comparison of Starting & Tuned Models

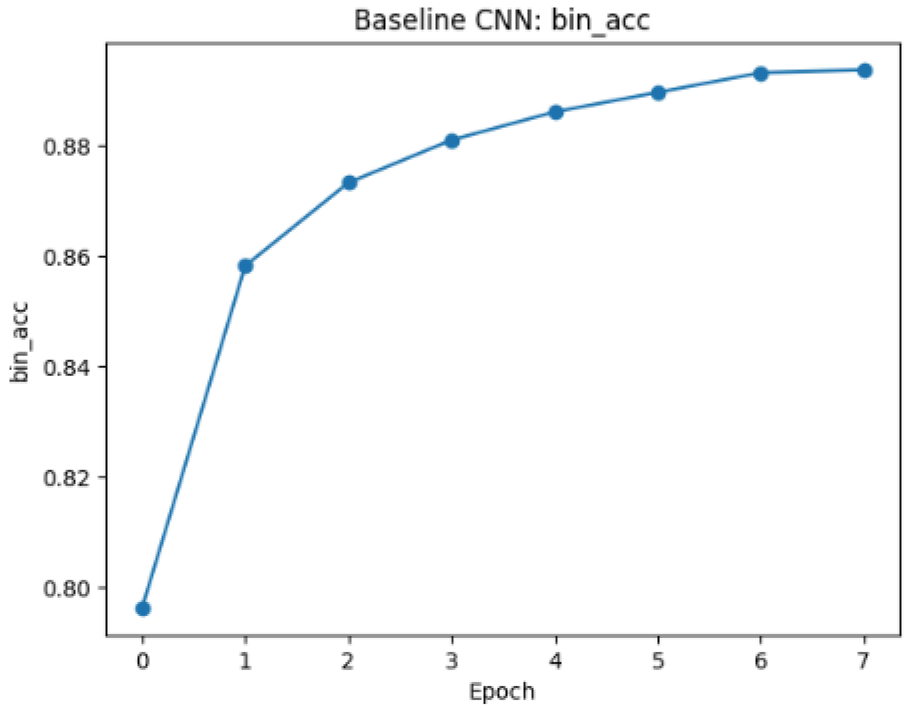
	label	f1_start	f1_tuned	f1_change
13	VALENTIA_pleasant_weather	0.014388	0.462366	0.447977
9	MUNCHENB_pleasant_weather	0.729505	0.811837	0.082333
12	KASSEL_pleasant_weather	0.676419	0.754811	0.078392
10	BASEL_pleasant_weather	0.690522	0.765525	0.075004
7	STOCKHOLM_pleasant_weather	0.761431	0.833686	0.072255
8	OSLO_pleasant_weather	0.736137	0.802938	0.066801
11	HEATHROW_pleasant_weather	0.681859	0.743005	0.061146
3	LJUBLJANA_pleasant_weather	0.799227	0.846055	0.046829
2	BUDAPEST_pleasant_weather	0.815544	0.850911	0.035367
5	DUSSELDORF_pleasant_weather	0.775825	0.807692	0.031867
1	BELGRADE_pleasant_weather	0.822692	0.851176	0.028485
6	DEBILT_pleasant_weather	0.765957	0.792214	0.026257
4	MAASTRICHT_pleasant_weather	0.784279	0.809481	0.025203
0	MADRID_pleasant_weather	0.897377	0.920830	0.023454
14	SONNBLICK_pleasant_weather	0.000000	0.000000	0.000000

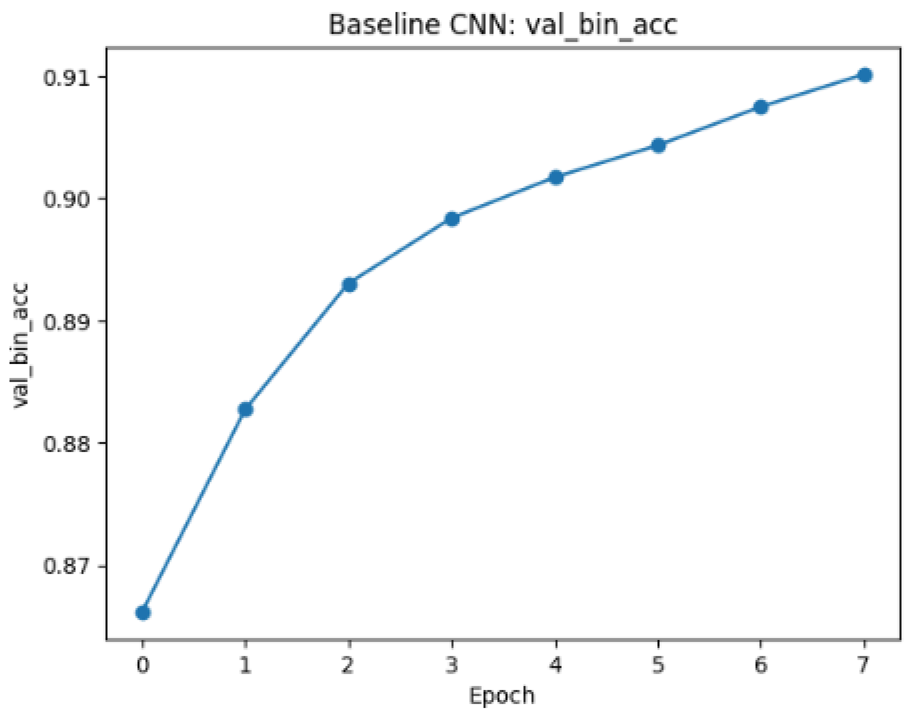
Hyperparameter tuning led to a clear and consistent improvement in overall performance, with macro-F1 increasing from ~0.66 to ~0.74 and macro-recall improving from ~0.65 to ~0.70. Nearly all stations show positive F1 gains, with the largest improvements observed for previously weaker labels such as Valentia, Munchen, and Kassel. These numbers also indicate improved generalization rather than overfitting to high-support stations. Performance for already strong stations remains stable or slightly improved, while Sonnblick shows no change due to the absence of positive samples in the test set. Overall, the tuned model demonstrates more balanced multi-label recognition across stations.

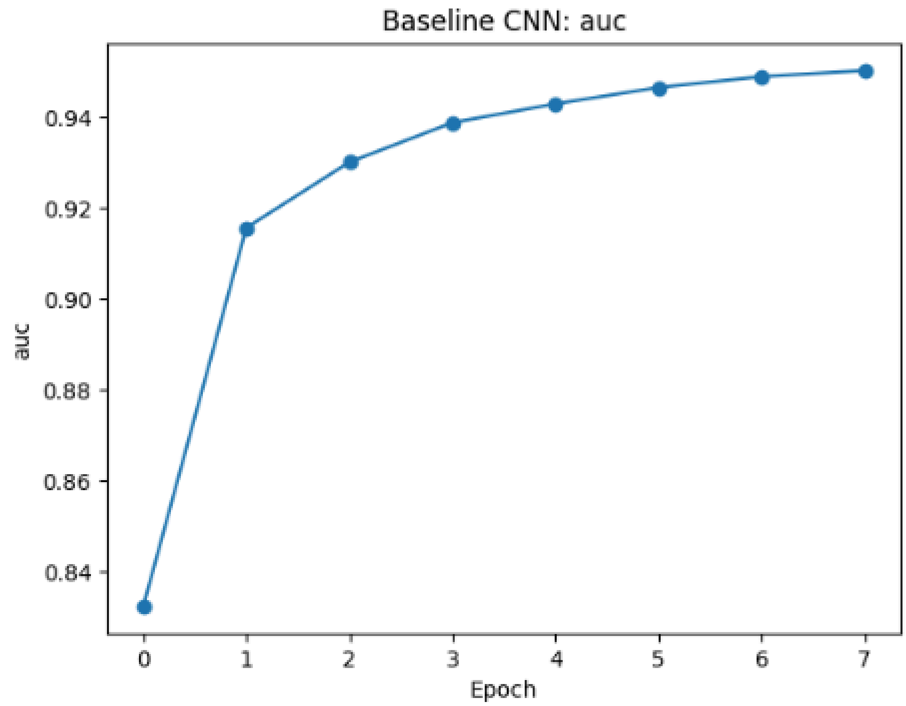
Learning Curve Plots

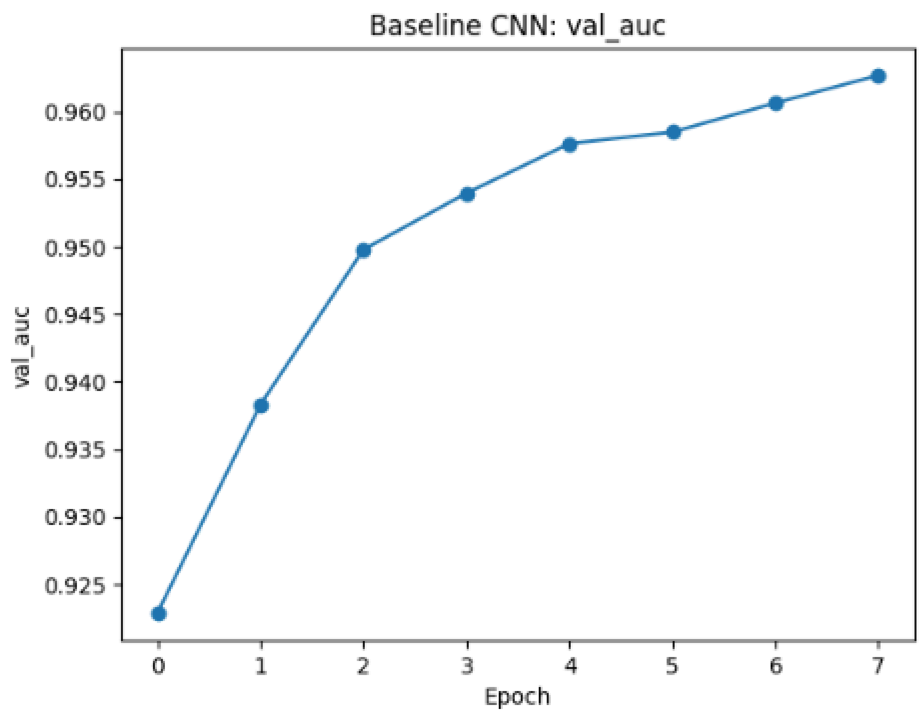


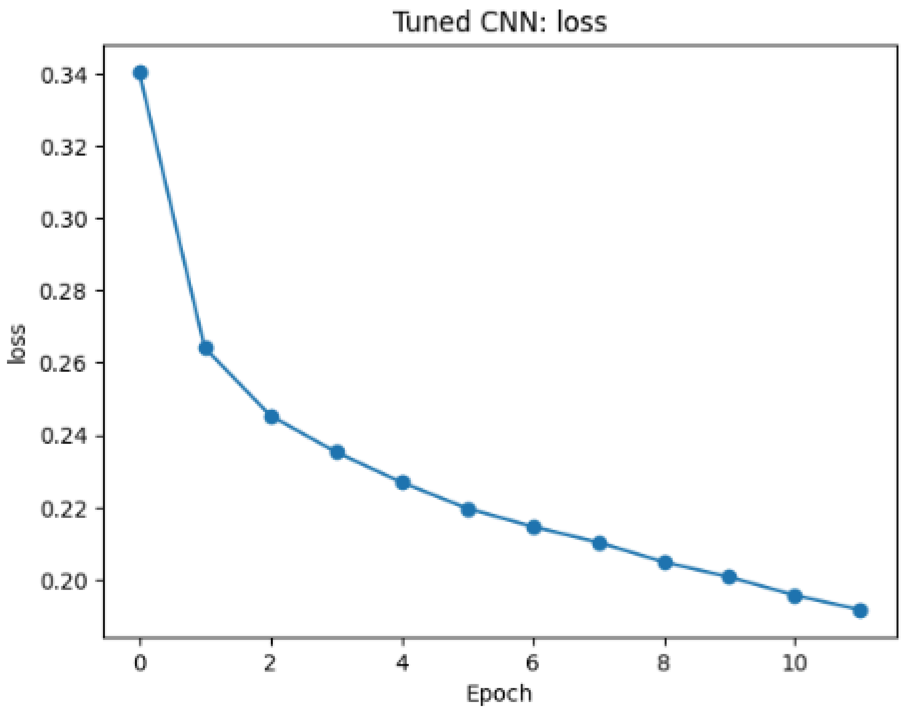


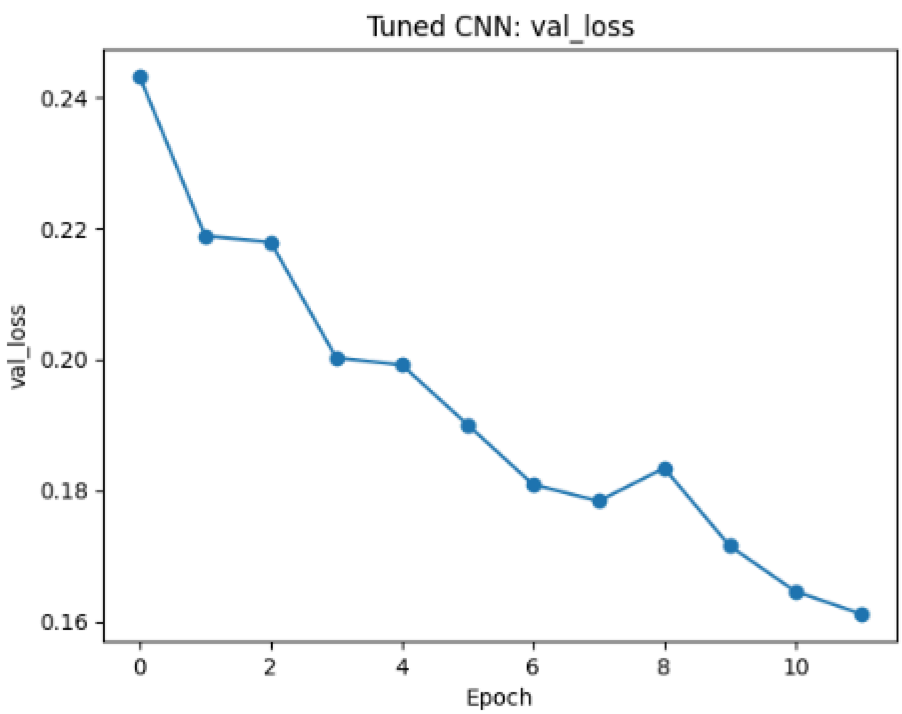

Baseline CNN: bin_acc
bin_acc
0.80
0.82
0.84
0.86
0.88
0
1
2
3
4
5
6
7
Epoch

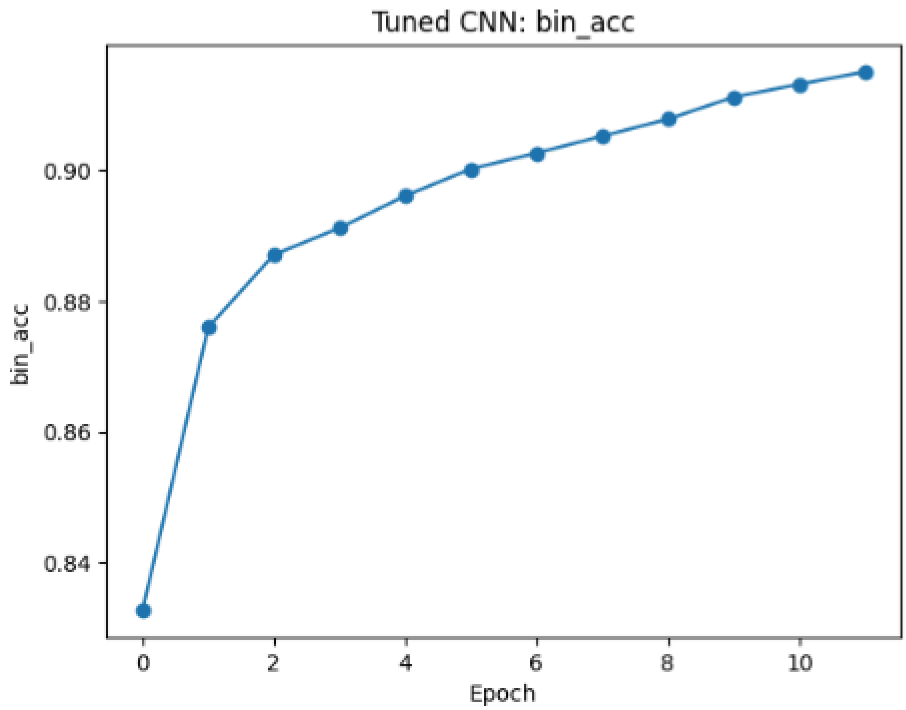

Baseline CNN: val_bin_acc
val_bin_acc
0.87
0.88
0.89
0.90
0.91
0
1
2
3
4
5
6
7
Epoch

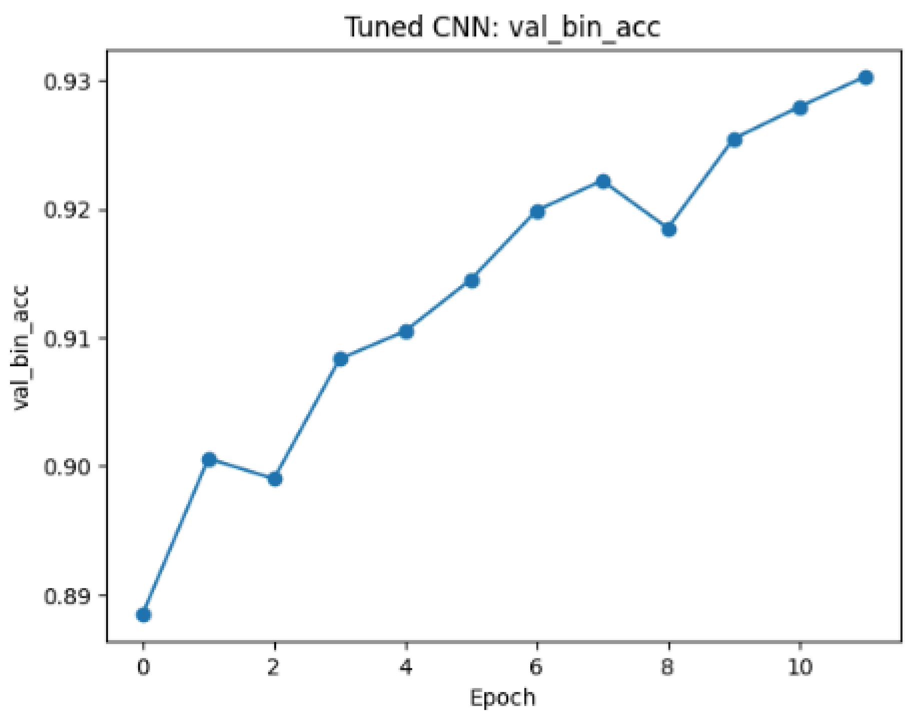

Baseline CNN: auc
auc
0.84
0.86
0.88
0.90
0.92
0.94
0 1 2 3 4 5 6 7
Epoch

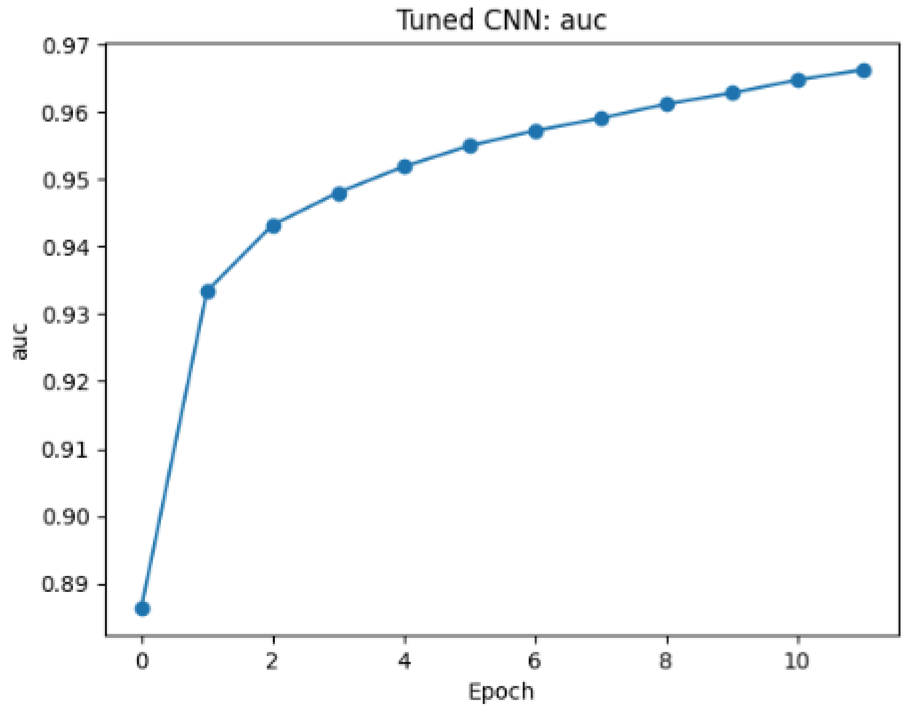

Baseline CNN: val_auc
val_auc
0.925
0.930
0.935
0.940
0.945
0.950
0.955
0.960
0 1 2 3 4 5 6 7
Epoch

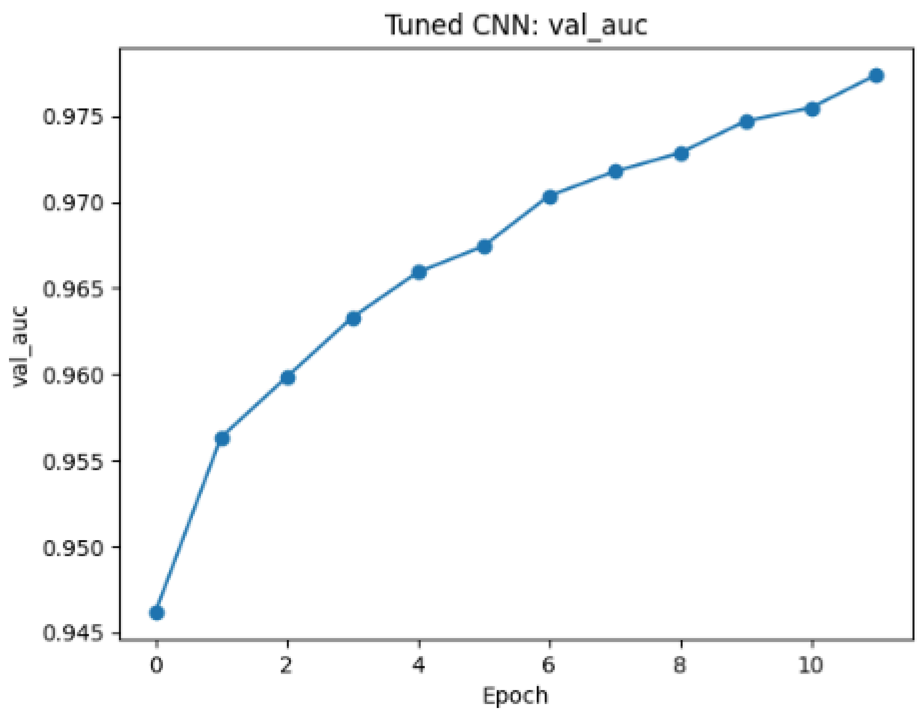

Tuned CNN: loss
loss
0.34
0.32
0.30
0.28
0.26
0.24
0.22
0.20
0
2
4
6
8
10
Epoch


Tuned CNN: val_loss
val_loss
0.24
0.22
0.20
0.18
0.16
0
2
4
6
8
10
Epoch


Tuned CNN: bin_acc
bin_acc
Epoch
0.84
0.86
0.88
0.90
0
2
4
6
8
10


Tuned CNN: val_bin_acc
val_bin_acc
Epoch
0.89
0.90
0.91
0.92
0.93
0
2
4
6
8
10


Tuned CNN: auc
auc
Epoch
0.97
0.96
0.95
0.94
0.93
0.92
0.91
0.90
0.89
0
2
4
6
8
10


Tuned CNN: val_auc
val_auc
Epoch
0.975
0.970
0.965
0.960
0.955
0.950
0.945
0
2
4
6
8
10

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Interpretation of the Plots

1. The model is learning smoothly

- Both baseline and tuned models show monotonic decreases in training and validation loss.
- No oscillation, no divergence, no late-epoch spikes.
- Validation loss consistently tracks *below or very close to* training loss.

This shows that the modeling has stable optimization.

2. No overfitting, and tuning helped

Baseline:	Tuned:
• Train AUC plateaus around ~0.95 • Validation AUC ~0.96	• Train AUC reaches ~0.97 • Validation AUC reaches ~0.978

Validation AUC is slightly higher than training AUC, which suggests that regularization and model capacity are effective.

3. Binary accuracy: steady, realistic improvement

- Baseline validation bin_acc ends ~0.91
- Tuned validation bin_acc ends ~0.93

The model is learning better decision boundaries. Gains are reasonable and incremental, and thresholding at 0.5 is sensible.

Additional Observations

Did loss decrease and accuracy/AUC increase?

Training results show a clear decrease in loss and corresponding increases in binary accuracy and AUC, indicating effective learning and stable convergence. The tuned CNN further improves performance without evidence of overfitting.

Can it recognize all 15 stations (i.e., do some labels have non-zero recall)?

The model successfully recognizes all stations that have positive examples in the dataset, with non-zero recall for 14 of 15 labels. One station exhibits zero recall due to a lack of positive samples, reflecting a data imbalance rather than a failure of the model.